

Project 6 : LED Glow

Introduction:

Pluto X2 can be added an external LED. LED module has more luminosity than the ones onboard. Here we will also, see how to use PWM capabilities present on pluto drone.

Challenge:

Build a program to glow the LED module steadily using Pluto flight controller board.

Add-On used:

A LED module is a simple LED attachable to Pluto drone. It can be used as an indicator or just a glowing-fading signal lamp.

Connection:

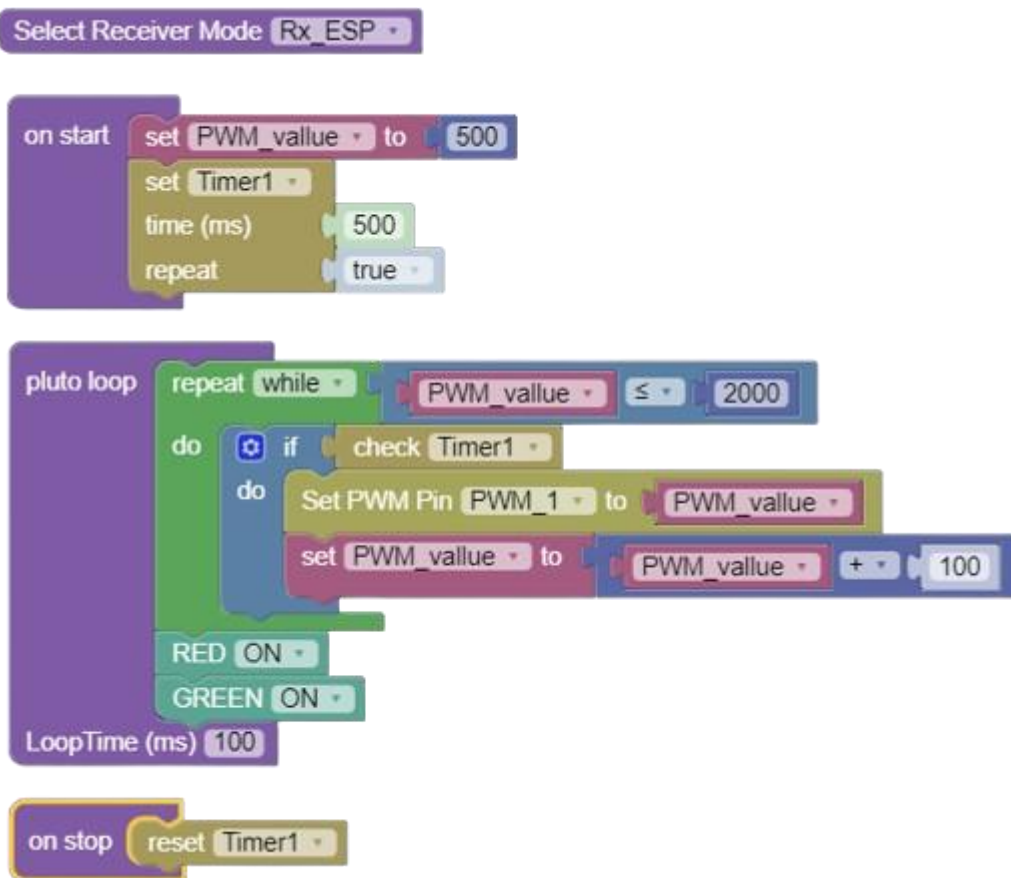
Connect the LED module on the RC port on Primus V5/X2 board

Primus V5/X2 pin	Peripherals
GND on RC port	GND on LED_module
Signal on RC port	VCC on LED_module

Approaching the Problem:

We will be using timer with $\frac{1}{2}$ second period and PWM functionality of Primus X2/V5 board. We will utilise while loop and manipulate PWM value going to LED, thus creating a glowing effect.

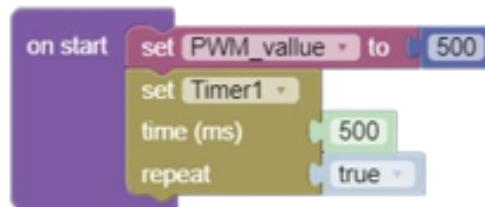
Solution:



Explanation of Solution:

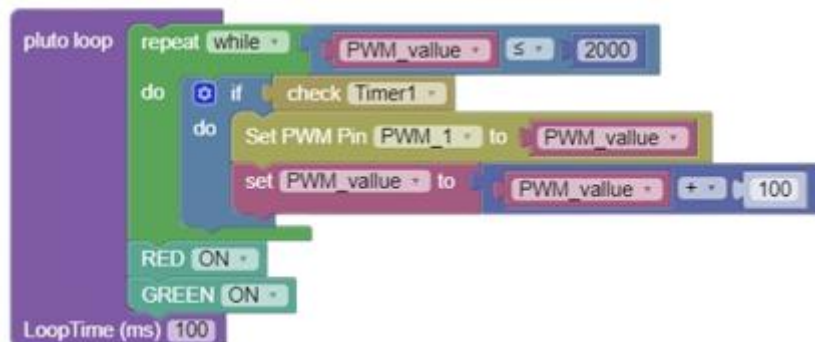
On Start Loop:

We set a repeatable timer with time of 500 milliseconds and a variable named PWM_value, given an initial value of 500.



Pluto Loop:

In PlutoLoop, a While loop runs that constantly provides LED with different PWM values from 500 to 2000 in an ascending order. Between two consecutive signals to LED, there is a delay of 500 milliseconds, provided by timer1. All the while, the Yellow light glows on Primus V5/X2 board.



On stop Loop: Timer1 resets.



Try further!

- Make use of different types of LEDs.
- Make a drone for search operations in tunnels/caves